

SQL Subqueries

Complete Notes with Practical Examples

Prepared for Students

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1 Introduction

A Subquery is a query written inside another SQL query.

A subquery can be used inside

- SELECT
- FROM
- WHERE
- HAVING
- INSERT
- UPDATE
- DELETE

Syntax

```
1 SELECT column_name
2
3 FROM table_name
4
5 WHERE column_name operator
6
7 (
8     SELECT column_name
9     FROM table_name
10 );
```

2 Sample Employee Table

Assume Employee table

ID	Name	Department	Salary
1	Aftab	IT	70000
2	Rahul	IT	80000
3	Priya	HR	60000
4	Mohit	HR	55000
5	Neha	Sales	90000
6	Aman	Sales	85000

3 Types of Subqueries

1. Single Row Subquery
2. Multiple Row Subquery
3. Multiple Column Subquery
4. Correlated Subquery
5. Nested Subquery

6. Scalar Subquery
7. Inline View
8. EXISTS
9. NOT EXISTS
10. ANY
11. ALL

4 Example 1 : Average Salary

Find employees whose salary is greater than average salary.

```
1 SELECT *
2
3 FROM Employee
4
5 WHERE Salary >
6
7 (
8 SELECT AVG(Salary)
9
10 FROM Employee
11 );
```

5 Example 2 : Maximum Salary

```
1 SELECT *
2
3 FROM Employee
4
5 WHERE Salary =
6
7 (
8 SELECT MAX(Salary)
9
10 FROM Employee
11 );
```

6 Example 3 : Minimum Salary

```
1 SELECT *
2
3 FROM Employee
4
5 WHERE Salary =
6
7 (
8 SELECT MIN(Salary)
9
10 FROM Employee
11 );
```

7 Example 4 : Highest Paid Employee

```
1 SELECT Name
2
3 FROM Employee
4
5 WHERE Salary=
6
7 (
8 SELECT MAX(Salary)
9
10 FROM Employee
11 );
```

8 Example 5 : Lowest Paid Employee

```
1 SELECT Name
2
3 FROM Employee
4
5 WHERE Salary=
6
7 (
8 SELECT MIN(Salary)
9
10 FROM Employee
11 );
```

9 Example 6 : Department Having High Salary

```
1 SELECT *
2
3 FROM Employee
4
5 WHERE Department IN
6
7 (
8 SELECT Department
9
10 FROM Employee
11
12 WHERE Salary>80000
13 );
```

10 Example 7 : Employees in HR Department

```
1 SELECT *
2
3 FROM Employee
4
5 WHERE Department=
```

```

6 |
7 | (
8 | SELECT Department
9 |
10 | FROM Employee
11 |
12 | WHERE Name='Priya'
13 | );

```

11 Example 8 : Scalar Subquery

Returns one value.

```

1 | SELECT
2 |
3 | Name ,
4 |
5 | Salary ,
6 |
7 | (
8 | SELECT AVG(Salary)
9 |
10 | FROM Employee
11 | )
12 |
13 | AS AverageSalary
14 |
15 | FROM Employee;

```

12 Example 9 : Subquery in SELECT

```

1 | SELECT
2 |
3 | Name ,
4 |
5 | (
6 | SELECT COUNT(*)
7 |
8 | FROM Employee
9 |
10 | )
11 |
12 | AS TotalEmployees
13 |
14 | FROM Employee;

```

13 Example 10 : Subquery in FROM

```

1 | SELECT *
2 |
3 | FROM
4 |
5 | (

```

```

6 SELECT Department ,
7
8 AVG(Salary)
9
10 AS AvgSalary
11
12 FROM Employee
13
14 GROUP BY Department
15
16 )
17
18 AS Temp;

```

14 Example 11 : Correlated Subquery

Executed once for every row.

```

1 SELECT *
2
3 FROM Employee E1
4
5 WHERE Salary >
6
7 (
8
9 SELECT AVG(Salary)
10
11 FROM Employee E2
12
13 WHERE E1.Department=E2.Department
14
15 );

```

15 Example 12 : EXISTS

```

1 SELECT *
2
3 FROM Employee E
4
5 WHERE EXISTS
6
7 (
8
9 SELECT *
10
11 FROM Employee
12
13 WHERE Department=E.Department
14
15 );

```

16 Example 13 : NOT EXISTS

```

1 SELECT *
2
3 FROM Employee E
4
5 WHERE NOT EXISTS
6
7 (
8 SELECT *
9
10 FROM Employee
11
12 WHERE Department='Marketing'
13 );

```

17 Example 14 : ANY

```

1 SELECT *
2
3 FROM Employee
4
5 WHERE Salary >
6
7 ANY
8
9 (
10 SELECT Salary
11
12 FROM Employee
13
14 WHERE Department='HR'
15 );

```

18 Example 15 : ALL

```

1 SELECT *
2
3 FROM Employee
4
5 WHERE Salary >
6
7 ALL
8
9 (
10 SELECT Salary
11
12 FROM Employee
13
14 WHERE Department='HR'
15 );

```

19 Example 16 : IN


```

1 SELECT *
2
3 FROM Employee
4
5 WHERE Department IN
6
7 (
8 SELECT Department
9
10 FROM Employee
11 );

```

20 Example 17 : NOT IN

```

1 SELECT *
2
3 FROM Employee
4
5 WHERE Department NOT IN
6
7 (
8 SELECT Department
9
10 FROM Employee
11
12 WHERE Salary > 80000
13 );

```

21 Example 18 : Second Highest Salary

```

1 SELECT MAX(Salary)
2
3 FROM Employee
4
5 WHERE Salary <
6
7 (
8 SELECT MAX(Salary)
9
10 FROM Employee
11
12
13 );

```

22 Example 19 : Third Highest Salary

```

1 SELECT MAX(Salary)
2
3 FROM Employee
4
5 WHERE Salary <

```

```

6
7 (
8
9 SELECT MAX(Salary)
10
11 FROM Employee
12
13 WHERE Salary<
14
15 (
16
17 SELECT MAX(Salary)
18
19 FROM Employee
20
21 )
22
23 );

```

23 Example 20 : Employee Above Department Average

```

1 SELECT *
2
3 FROM Employee E1
4
5 WHERE Salary>
6
7 (
8
9 SELECT AVG(Salary)
10
11 FROM Employee E2
12
13 WHERE E1.Department=E2.Department
14
15 );

```

24 Example 21 : Count Employees

```

1 SELECT
2
3 Department ,
4
5 (
6
7 SELECT COUNT(*)
8
9 FROM Employee
10
11 )
12
13 AS Total
14
15 FROM Employee;

```

25 Example 22 : Update Using Subquery

```
1 UPDATE Employee
2
3 SET Salary=
4
5 Salary+5000
6
7 WHERE Department=
8
9 (
10
11 SELECT Department
12
13 FROM Employee
14
15 WHERE Name= 'Rahul '
16
17 );
```

26 Example 23 : Delete Using Subquery

```
1 DELETE FROM Employee
2
3 WHERE Department=
4
5 (
6
7 SELECT Department
8
9 FROM Employee
10
11 WHERE Name= 'Mohit '
12
13 );
```

27 Example 24 : Insert Using Subquery

```
1 INSERT INTO Employee_Backup
2
3 SELECT *
4
5 FROM Employee
6
7 WHERE Salary>
8
9 (
10
11 SELECT AVG(Salary)
12
13 FROM Employee
14
15 );
```

28 Example 25 : Nested Subquery

```
1 SELECT *
2
3 FROM Employee
4
5 WHERE Department=
6
7 (
8
9 SELECT Department
10
11 FROM Employee
12
13 WHERE Salary=
14
15 (
16
17 SELECT MAX(Salary)
18
19 FROM Employee
20
21 )
22
23 );
```

29 Difference Between IN and EXISTS

IN	EXISTS
Compares values	Checks existence
Returns list	Returns TRUE/FALSE
Suitable for small datasets	Better for large datasets
Usually slower on huge tables	Usually faster

30 Difference Between ANY and ALL

ANY	ALL
One condition must satisfy	All conditions must satisfy
Less restrictive	More restrictive

31 Interview Questions

1. What is a Subquery?
2. Types of Subqueries?
3. Difference between Correlated and Non-Correlated Subquery?
4. Difference between EXISTS and IN?
5. Difference between ANY and ALL?
6. Can Subquery return multiple rows?

7. What is Scalar Subquery?
8. Can we use Subquery inside SELECT?
9. Can we use Subquery inside UPDATE?
10. Difference between JOIN and Subquery?
11. What is Nested Subquery?
12. Which is faster JOIN or Subquery?

32 Summary

Subqueries are one of the most powerful SQL concepts. They allow one query to depend on the result of another query, making it possible to solve complex problems such as filtering, aggregation, ranking, comparisons, updates, and reporting without writing multiple separate SQL statements.